

<https://github.com/ahmedabdelmongy/ahmedabdelmongy-/blob/main/Machine-Learning.html>

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End of Module Assignment: e-Portfolio Submission

Module: Machine Learning

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E-Portfolio Link

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1. OVERVIEW

This e-portfolio presents evidence of my independent work across the Machine Learning module. It highlights my contributions, development of the final project, and structured reflection using the Rolfe model.

The portfolio focuses on building clear, reproducible workflows supported by results, figures, and code. Graphs and tables support the analysis, and all artefacts are linked through the GitHub repository. Writing follows Harvard referencing style and connects learning to real-world applications.

2. REFLECTIVE

2.1 WHAT

I developed a machine learning pipeline starting from preprocessing to evaluation. A baseline model was created first, followed by comparison with a tree-based approach.

Key steps included:

- cleaning features
- selecting appropriate metrics (MAE, R^2)
- simplifying the workflow
- creating clear visualisations

The final pipeline was structured for clarity, testing, and reproducibility.

2.2 SO WHAT

This improved my understanding of:

- model validation and data splitting
- avoiding data leakage

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- building strong baselines
- interpreting results clearly

I also improved communication through structured updates and clearer reporting using visualisations.

2.3 NOW WHAT

Next, I will:

- apply SHAP for model explainability
- compare with GAM models
- improve reproducibility using structured scripts
- enhance reporting clarity with concise summaries

2.4 Professional Skills Matrix

Skill Area	Baseline	Evidence	Improvement	Next Action
-----	-----	-----	-----	-----
Regression	3	Report & plots	Clear metrics	Apply SHAP
Validation	3	Final runs	Better splits	Checklist
Visualisation	2	Figures	Clear plots	Peer review
Collaboration	4	Notes	Better tracking	Shared tools
Writing	3	Feedback	Clear structure	Harvard checks

3. EVIDENCE OF LEARNING AND CONTRIBUTION

I implemented a structured modelling pipeline with:

- proper train/validation/test splits
- controlled random seeds
- baseline comparison

Metrics used:

- MAE

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- R^2

Figures and tables supported the analysis, ensuring clarity and reproducibility.

Unit 6 Contribution

I built the regression baseline and compared it with a tree model. Metrics were aligned with the problem, and results were visualised clearly.

Figure 1. MAE comparison between models (Linear, Ridge, Random Forest).

Unit 11 Contribution

I refined the pipeline by:

- improving validation
- removing noisy features
- aligning seeds across runs

Figure 2. Feature importance from tree model.

Figure 3. Training vs validation learning curve.

4. EVALUATION: FINAL PROJECT VS ORIGINAL PROPOSAL

The final project improved stability by:

- reducing feature complexity
- improving validation techniques
- focusing on clear metrics

Trade-offs:

- slightly lower accuracy
- higher reliability and interpretability

The pipeline became:

- easier to maintain
- more reproducible
- clearer for reviewers

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Problem – Approach – Impact – Risk – Next Steps

Problem:

Initial approach had too many features and unstable results.

Approach:

- simplify inputs
- standardise preprocessing
- use fixed seeds and fair comparisons

Impact:

- improved stability
- clearer results
- faster review

Risk:

- small drop in accuracy
- limited feature richness

Next Steps:

- SHAP explainability
- GAM comparison
- improved drift checks

5. PRFAQ (PRODUCT REVIEW Q&A)

Q: What changed from the proposal?

A: Reduced features and simplified models to improve stability.

Q: Why this model choice?

A: Linear for clarity, tree for nonlinear patterns.

Q: How were comparisons made fair?

A: Same seeds, splits, and repeated runs.

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Q: Which metrics matter?

A: MAE and R^2 for interpretability.

Q: What improved?

A: Stability, clarity, and reproducibility.

Q: What are limitations?

A: Limited features and need for explainability tools.

Q: How to improve?

A: Use SHAP and GAM models.

6. REFERENCES

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